COLLEGE OF MEDICINE

Approach To Normal And Abnormal Development Of Genital Tract

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LEARNING OBJECTIVES

- Understand that sexual differentiation and development begin in early embryonic life.

- Understand the embryonic development and the anatomy of the perineum, the vagina, cervix and uterus, the adnexa and ovary and the bladder and ureters.

- Describe the structural anomalies resulting from Müllerian tract disorders.

Sexual differentiation of the fetus and development of sexual organs:

- The chromosomal complement of the zygote determines whether the gonad becomes a testis or an ovary (bipotential).

- There are four main phases of genital development:

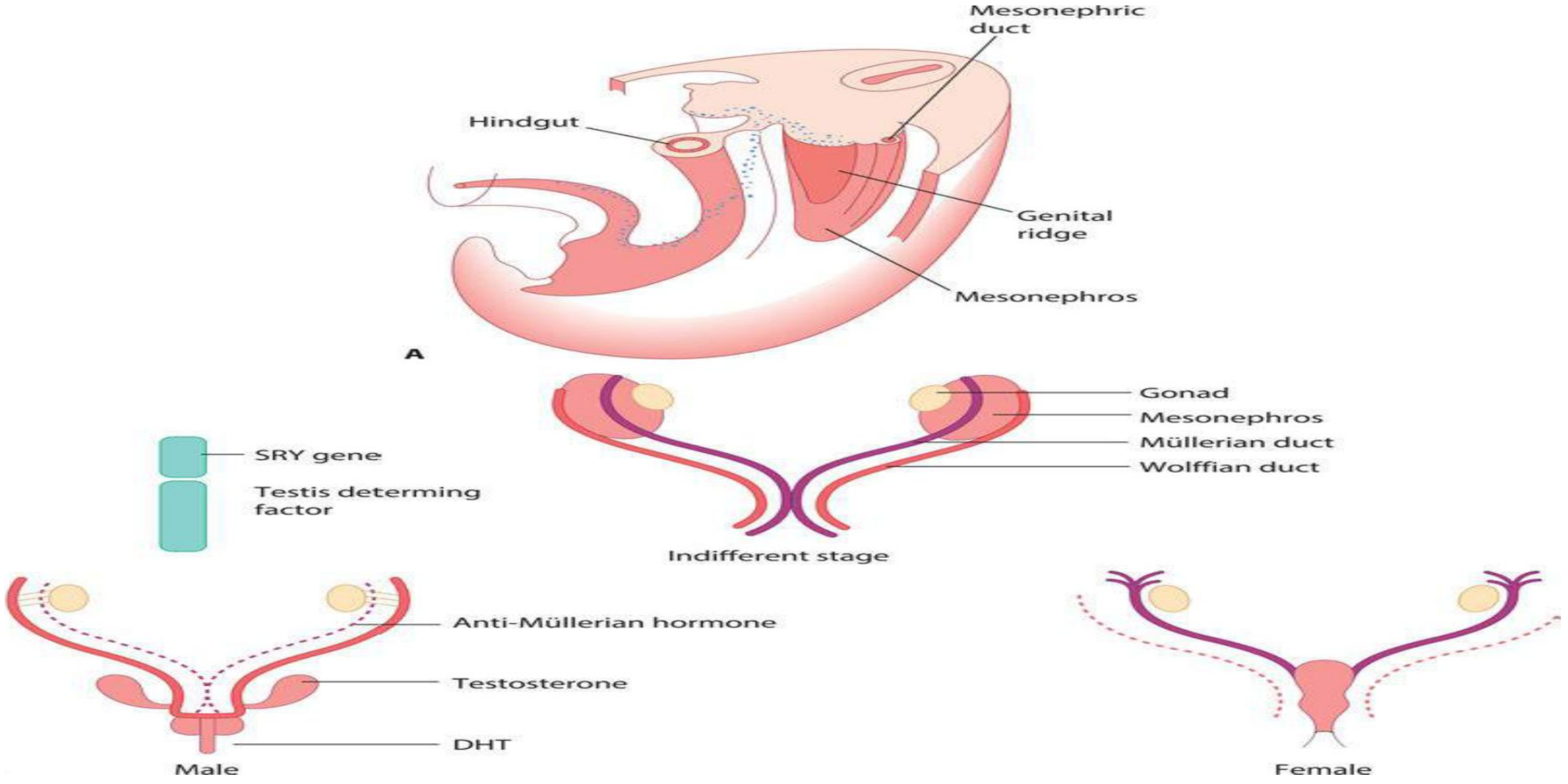
-the indifferent gonadal phase at 4-6 wks of gestation.

-the gonadal differentiation phase at 7 wk gestation.

- the ductal differentiation at 9-11 wks gestation.

-the external genitalia differentiation at 10-12 wks.

- The development of either the testis or ovary is an active gene-directed process.


Mesonephric duct
Hindgut
Genital ridge
Mesonephros
A
Gonad
Mesonephros
Müllerian duct
Wolffian duct
SRY gene
Testis determing factor
Indifferent stage
Anti-Müllerian hormone
Testosterone
DHT
Male
Female

- In the male the activity of the SRY gene (sex-determining region of the Y chromosome) causes the gonad to begin development into a testis.

- The fetus has two sets of structures called the Müllerian (or paramesonephric) ducts and Wolffian (or mesonephric) ducts, which have the potential to develop into female or male internal and external genitalia respectively.

Development of the male sexual organs:

- Testis differentiates into two cell types(Sertoli and lyedigs).

- The Sertoli cells produce antiMüllerian hormone (AMH) that suppresses further development of mullerian ducts.

- The Leydig cells produce testosterone that stimulates the Wolffian ducts to develop into the vas deferens, epididymis and seminal vesicles.

- The testosterone is converted by the enzyme 5-alpha-reductase into dihydrotestosterone (DHT) this will happen in the external genitalia in order to virilize this external genitalia.

- The genital tubercle becomes the penis and the labioscrotal folds fuse to form the scrotum.

- The urogenital folds fuse along the ventral surface of the penis and enclose the urethra so that it opens at the tip of the penis.

Development of the female sexual organs:

- In the primitive ovary granulosa cells, derived from the proliferating coelomic epithelium, surround the germ cells and form primordial follicles.

- Each primordial follicle consists of an oocyte within a single layer of granulosa cells, that septate from the theca cells by basal lamina.

- The number of primordial follicles is reached at 20 weeks' gestation when there are 6-7 million.

- Due to atresia, at birth only 1–2 million remain.
- by menarche only 300,000–400,000 are present, and by menopause none.

- The development of an oocyte within a primordial follicle is arrested at the prophase of its first meiotic division. It remains in that state until it undergoes atresia or enters the meiotic process preceding ovulation.

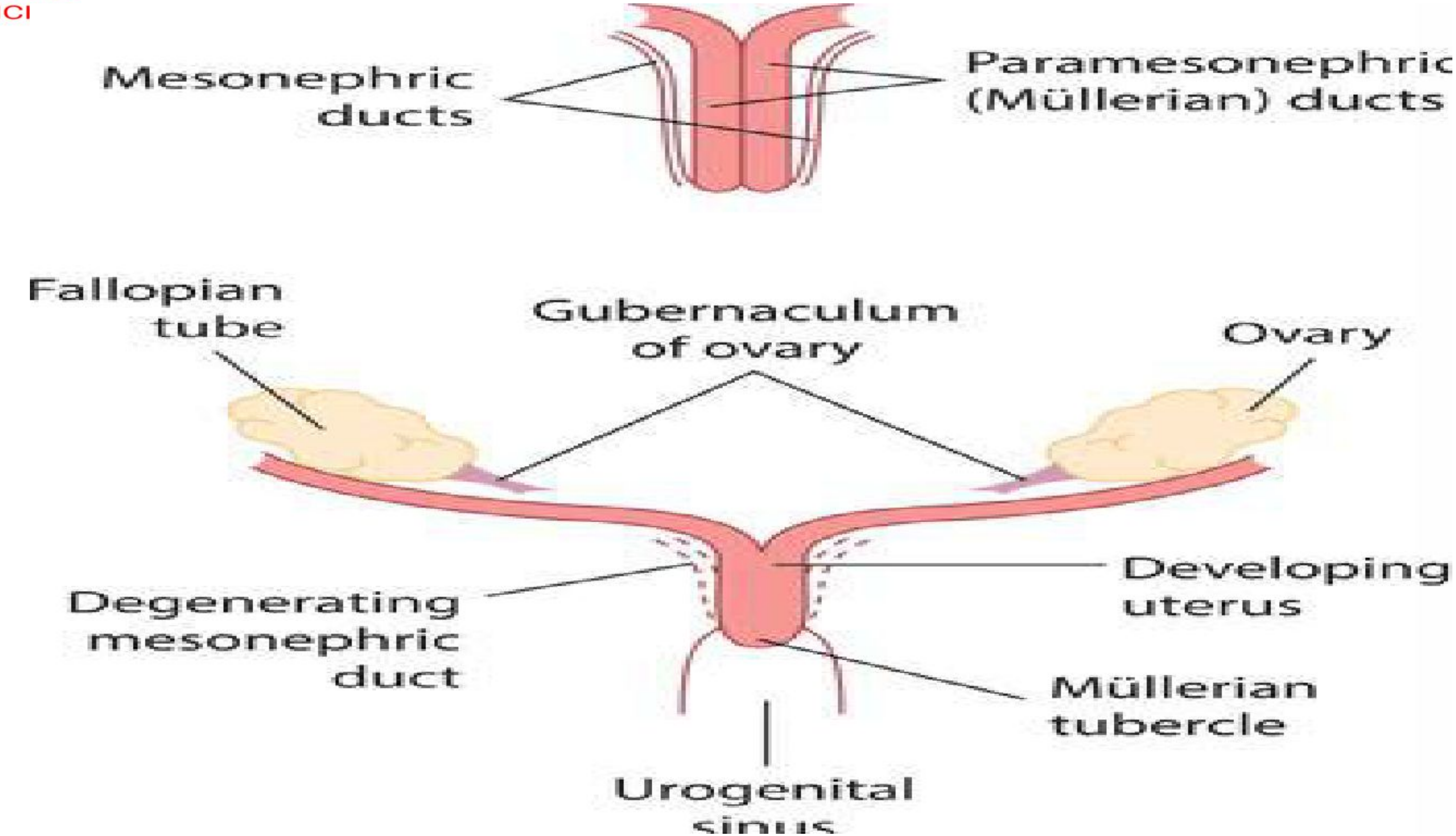
- The absence of testicular AMH allows the Müllerian structures to develop and the female reproductive tract develops from these paired ducts.

- At 9-11 wks the proximal two-thirds of the vagina develop from the paired Müllerian ducts, which grow in a caudal and medial direction and fuse in the midline (formation of the septum).

- The midline fusion of these structures produces the uterus, cervix and upper vagina, and the unfused caudal segments form the Fallopian tubes.

- Cells proliferate from the upper portion of the urogenital sinus to form structures called the 'sinovaginal bulbs'.

- The Müllerian tubercles and the urogenital sinus fuse to form the vaginal plate, which extends from the Müllerian ducts to the urogenital sinus.
- This plate begins to canalize, starting at the hymen and proceeding upwards to the cervix in the sixth embryonic month.
- At 5^{th} months , Mullerian organogenesis is complete with uterine septal resorption /canalization.



Caudal parts of the paramesonephric ducts (top) fuse to form the uterus and Fallopian tubes.

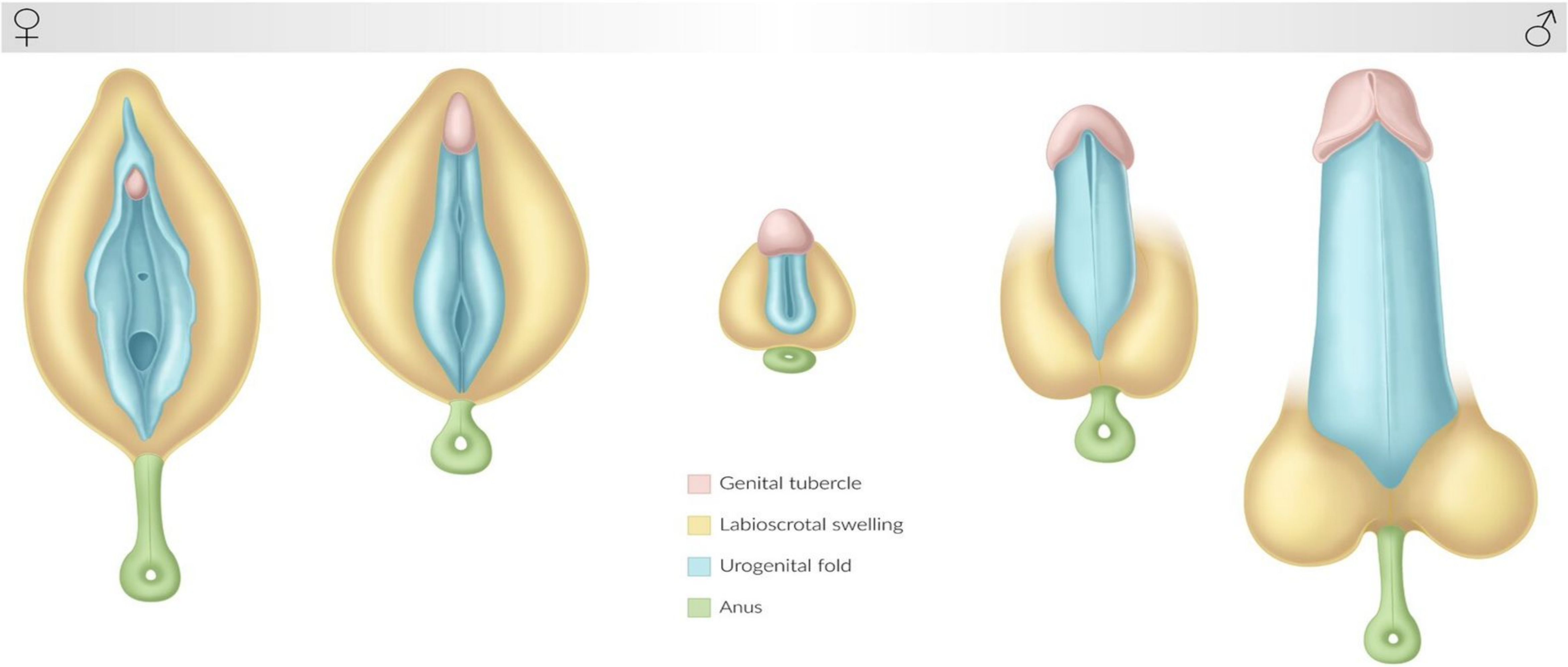
External female genitalia :

- The external genitalia don’t virilize in the absence of testosterone.
- Between 5th and 7th wks the cloacal folds, which are folds of swellings adjacent to the cloacal membranes , fuse anteriorly to become the genital tubercle. This will become the clitoris.
- The perineum will develops and divides the cloacal membrane into an anterior urogenital membrane and posterior anal membrane.
- The cloacal folds anteriorly are called urethral folds, which form the labia minora

- Another pairs of folds within the cloacal membranes form the labioscrotal folds that eventually become the labia majora.

- The urogenital sinus become the vestibule of the vagina.

- The external genitalia are recognizably female at the end of the 12^{th} embryonic wk.


♀
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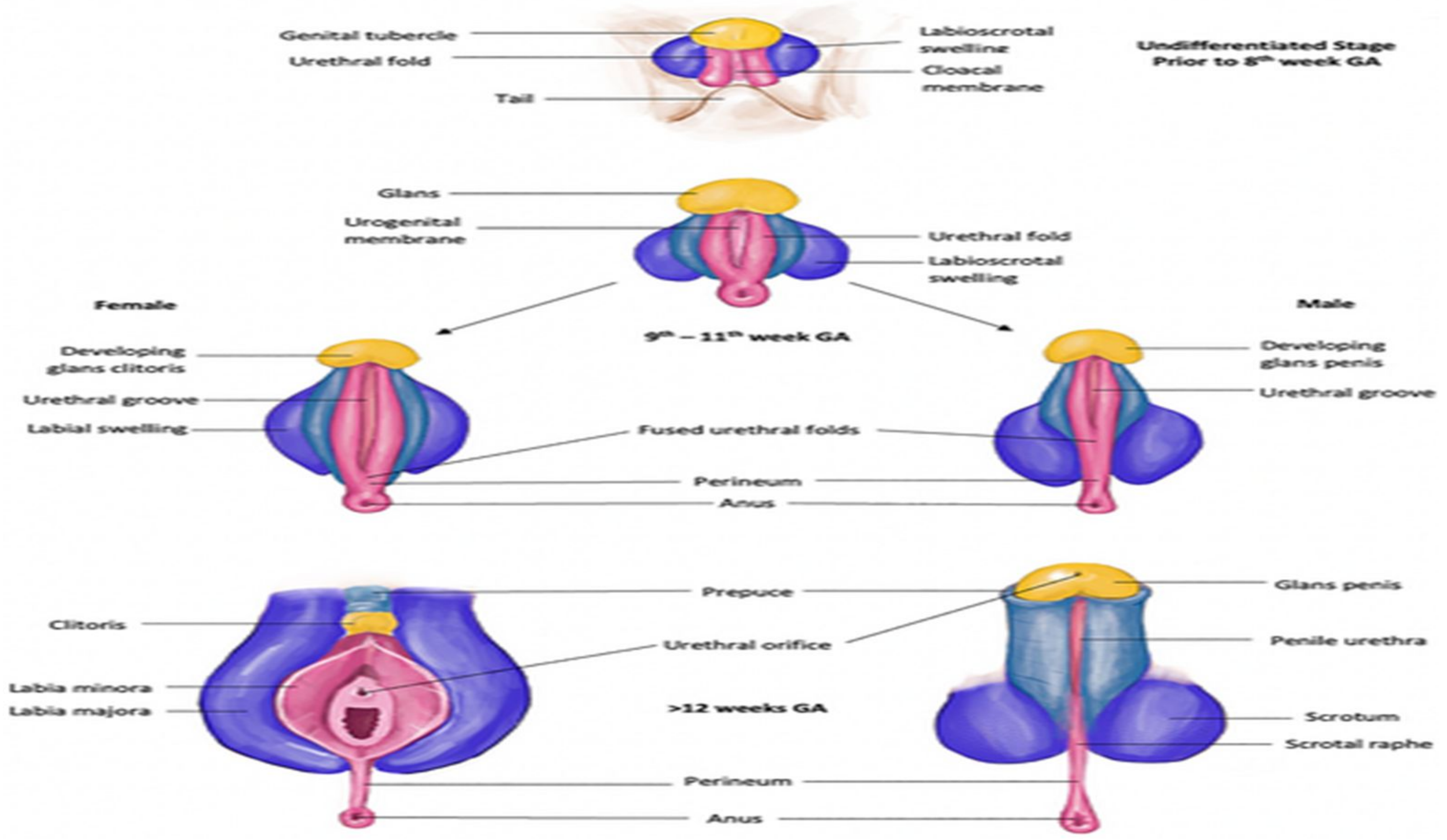
Genital tubercle
Labioscrotal swelling
Urogenital fold
Anus


Genital tubercle
Urethral fold
Tail
Labioscrotal swelling
Cloacal membrane
Undifferentiated Stage Prior to 8th week GA
Glans
Urogenital membrane
Urethral fold
Labioscrotal swelling
Female
Male
9th – 11th week GA
Developing glans clitoris
Urethral groove
Labial swelling
Fused urethral folds
Perineum
Anus
Developing glans penis
Urethral groove
Prepuce
Clitoris
Urethral orifice
Labia minora
Labia majora
>12 weeks GA
Perineum
Anus
Glans penis
Penile urethra
Scrotum
Scrotal raphe

KEY LEARNING POINTS :

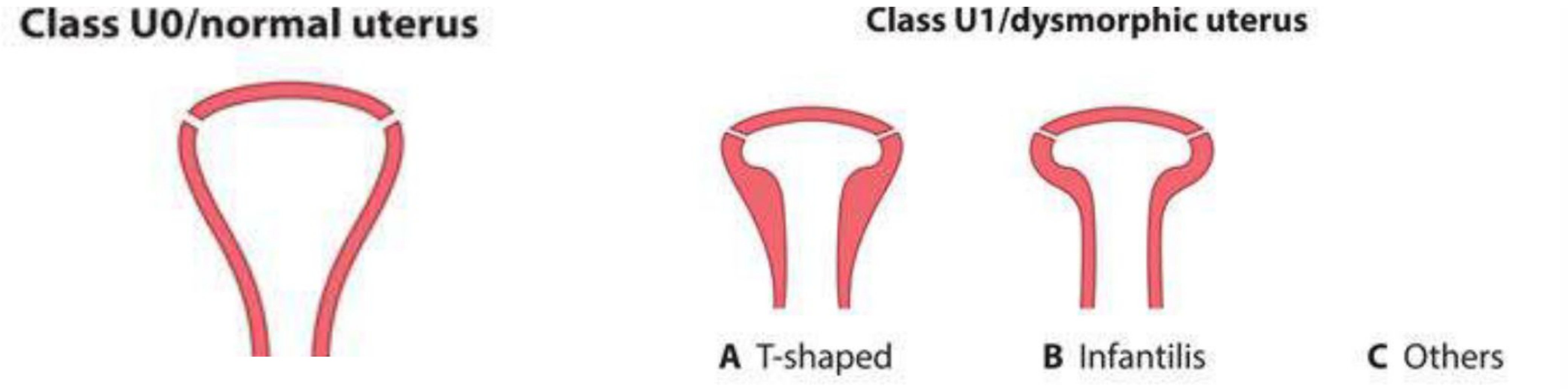
- The primitive gonad is first evident at 5 weeks of embryonic life and forms on the medial aspect of the mesonephric ridge.
- The undifferentiated gonad has the potential to become either a testis or an ovary.
- The paramesonephric duct, which later forms the Müllerian system, is the precursor of female genital development.
- The lower end of the Müllerian ducts fuse in the midline to form the uterus and upper vagina.
- Most of the upper vagina is of Müllerian origin, while the lower vagina forms from the sinovaginal bulbs.

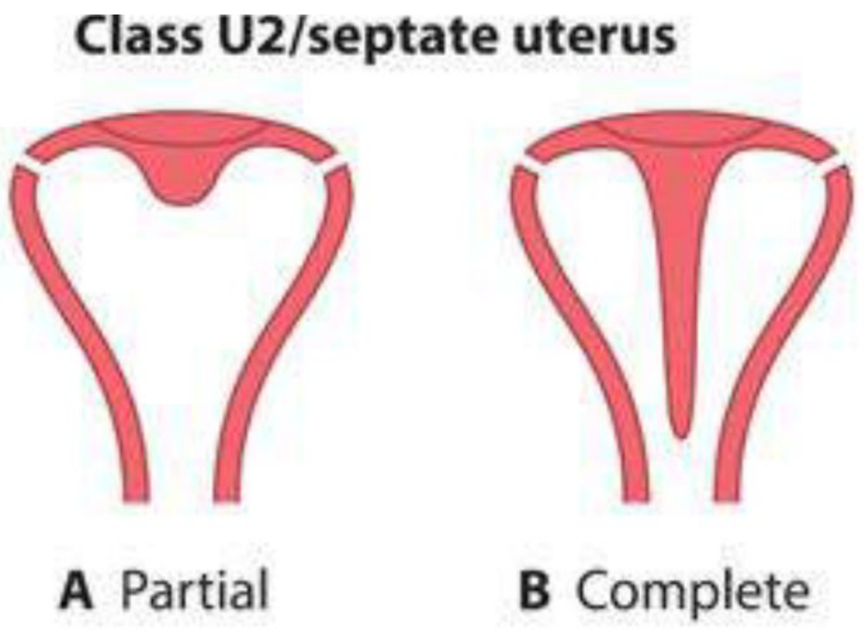
Structural problems of pelvic organs (Müllerian anomalies):

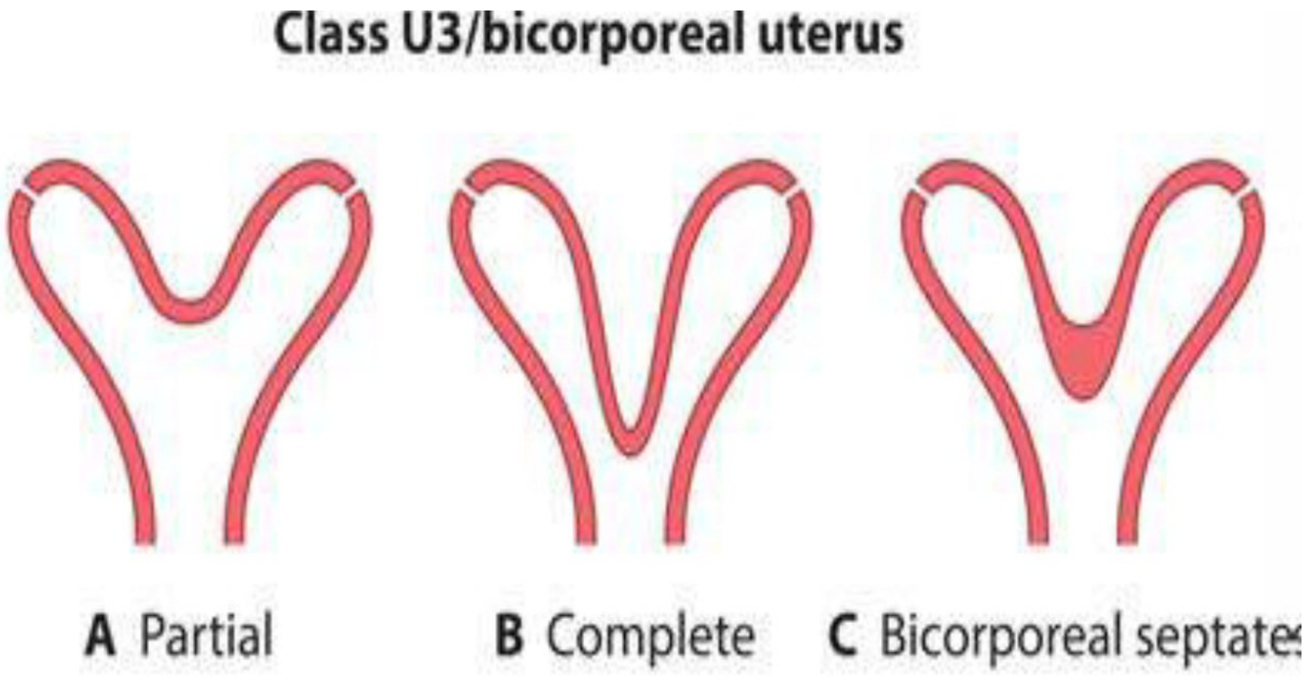
- Are common occurring in up to 6% of the female population, and may be asymptomatic.
- The aetiology is unknown, although associated renal anomalies are present in up to 30%.
- The clinical significances of the mullerian anomalies are shown the following table

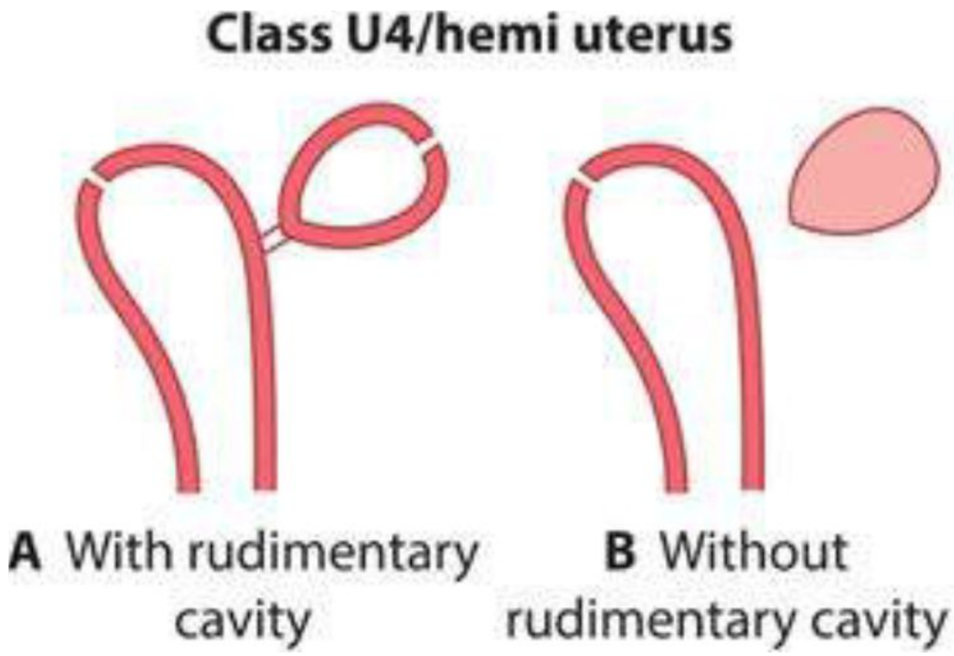
Mulllerian development	*Clinical significance*
1-organogesis	Defects leads to agenesis or hypoplasia(unicornuate or absent uterus)
2-fusion of the mullerian duct	Horizintalfusion or unification defect: • Partial-bicornuate ut • Complete-ut didelephys • Vertical fusion-imperforaye hymen or transverse vaginal septum
3- septal defect	Canalization defects: • Complete or partial septate ut • Arcuate ut

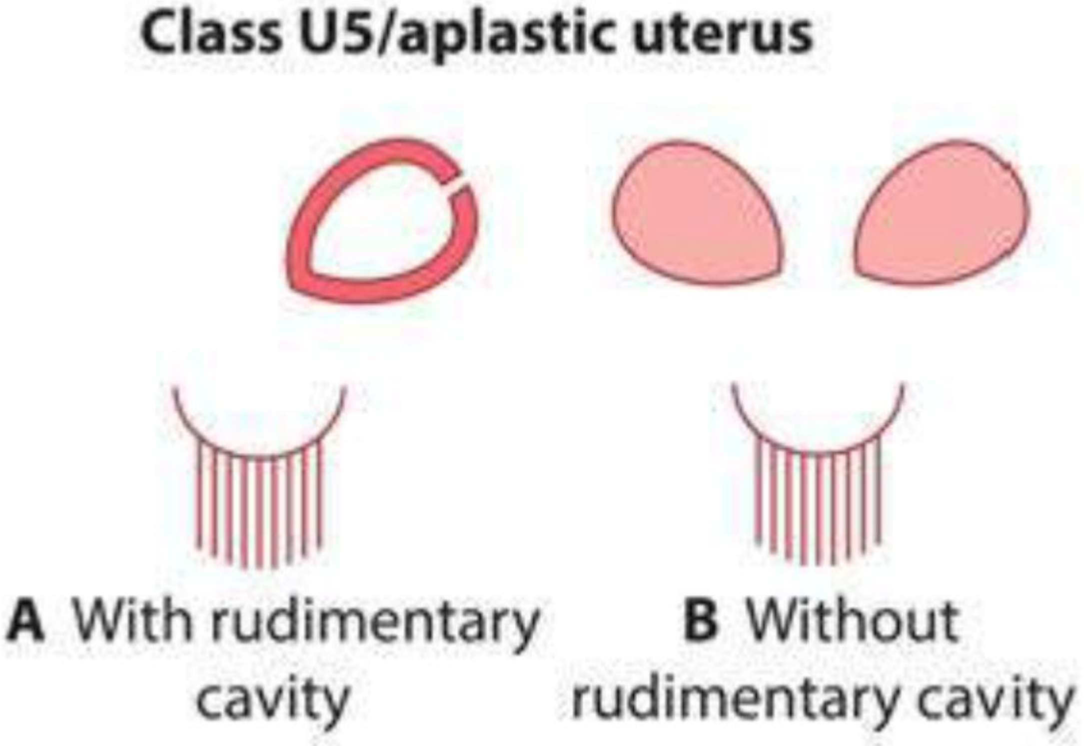
- The classification European for these anomalies as follow :




Class U2/septate uterus
A Partial
B Complete


Class U3/bicorporeal uterus
A Partial
B Complete
C Bicorporeal septate

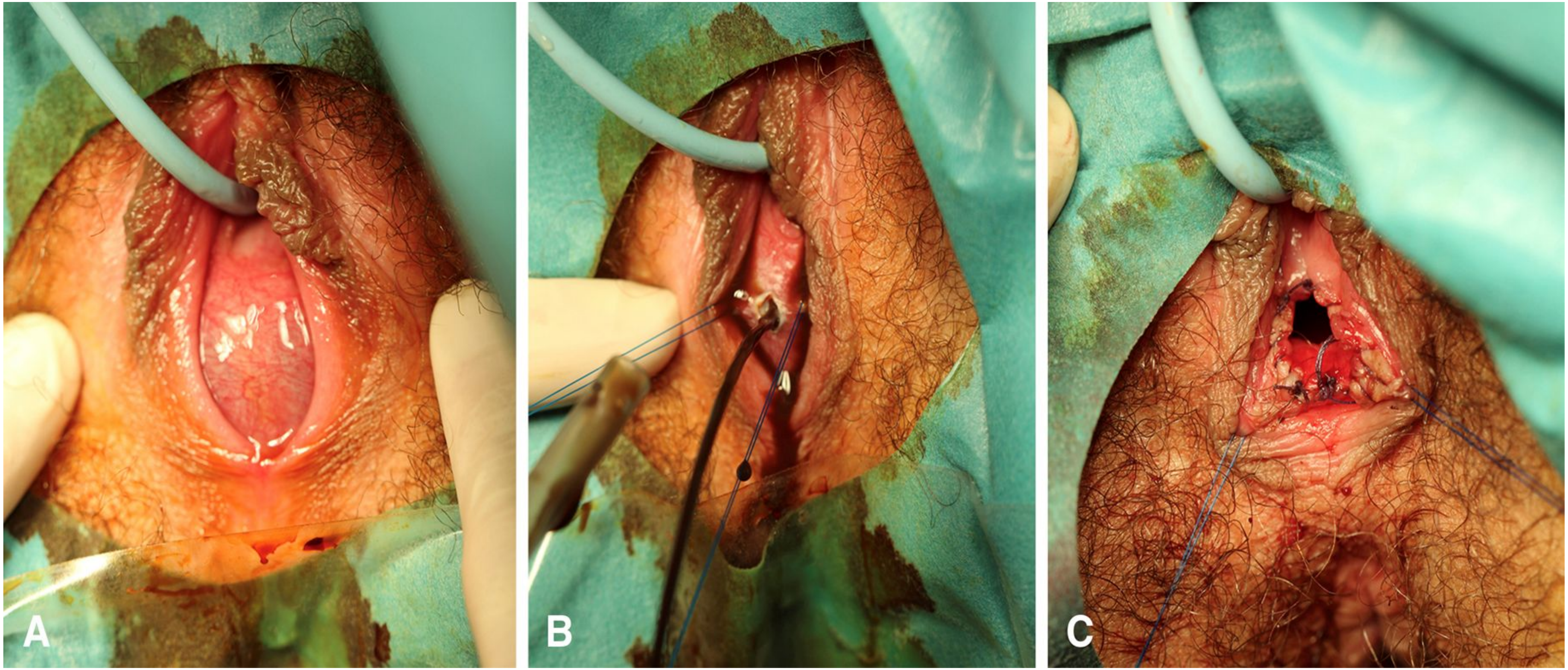




Class U6/unclassified cases

Müllerian obstruction

- lead to menstrual obstruction.
- The obstruction most commonly occurs at the junction of the lower third of the vagina at the level of the hymen the presentation cab be increasing abdominal pain in a girl in early adolescence.
- The retained menstrual blood stretches the vagina, causing a haematocolpus, large pelvic mass
- Treatment is simple with a surgical incision of the hymen and drainage of the retained blood.


A
B
C

Müllerian duplication :

- It may be a complete duplication of the uterus, cervix and vagina, but may be simply a midline uterine septum in otherwise normal internal genitalia.
- Second uterine horns may also occur and can be rudimentary or functional.

Müllerian agenesis:

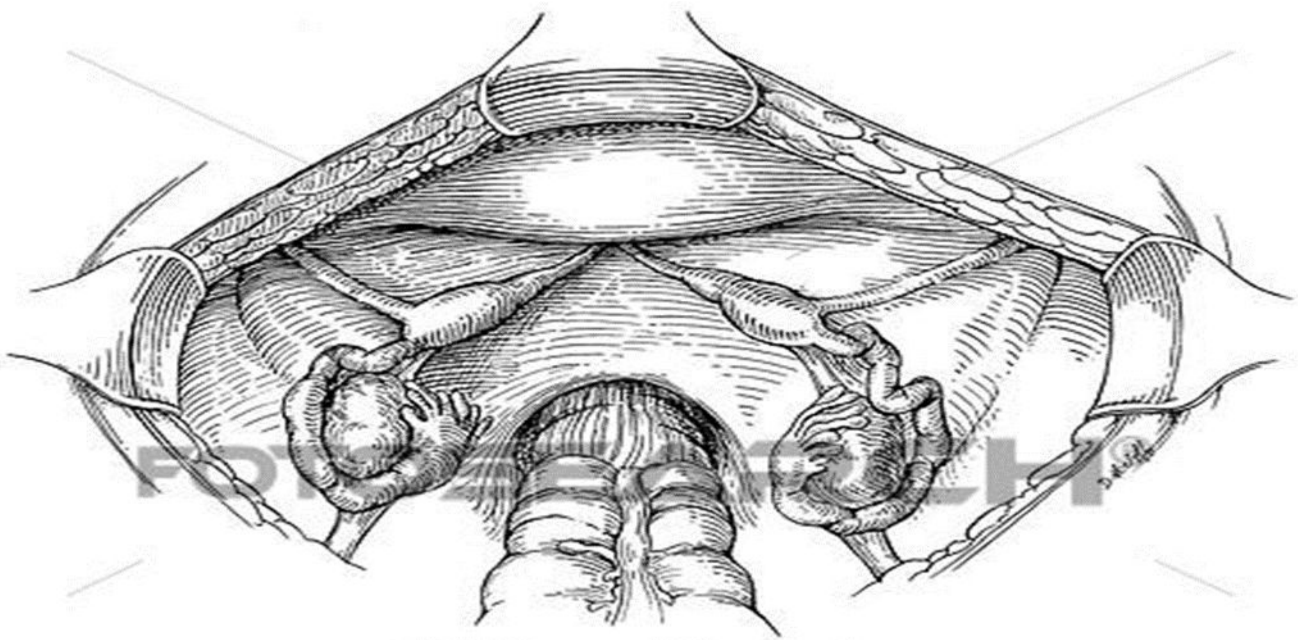
- In approximately 1 in 5,000 to 1 in 40,000 girls.
- resulting in an absent or rudimentary uterus and upper vagina.
- This condition is known as Rokitansky syndrome or Mayer–Rokitansky–Kuster–Hauser syndrome (MRKH).
- The ovaries function normally (primary amenorrhoea , otherwise normal pubertal development).

- The aetiology of this condition is not known,.

- the vagina will be blind ending and is likely to be shortened in length.
- An ultrasound scan will confirm the presence of ovaries, but no functioning uterus will be present.

- Treatment :psychological support and on the creation of a vagina comfortable for penetrative intercourse.

- may have their own genetic children, using ovum retrieval and assisted conception techniques, and a surrogate mother.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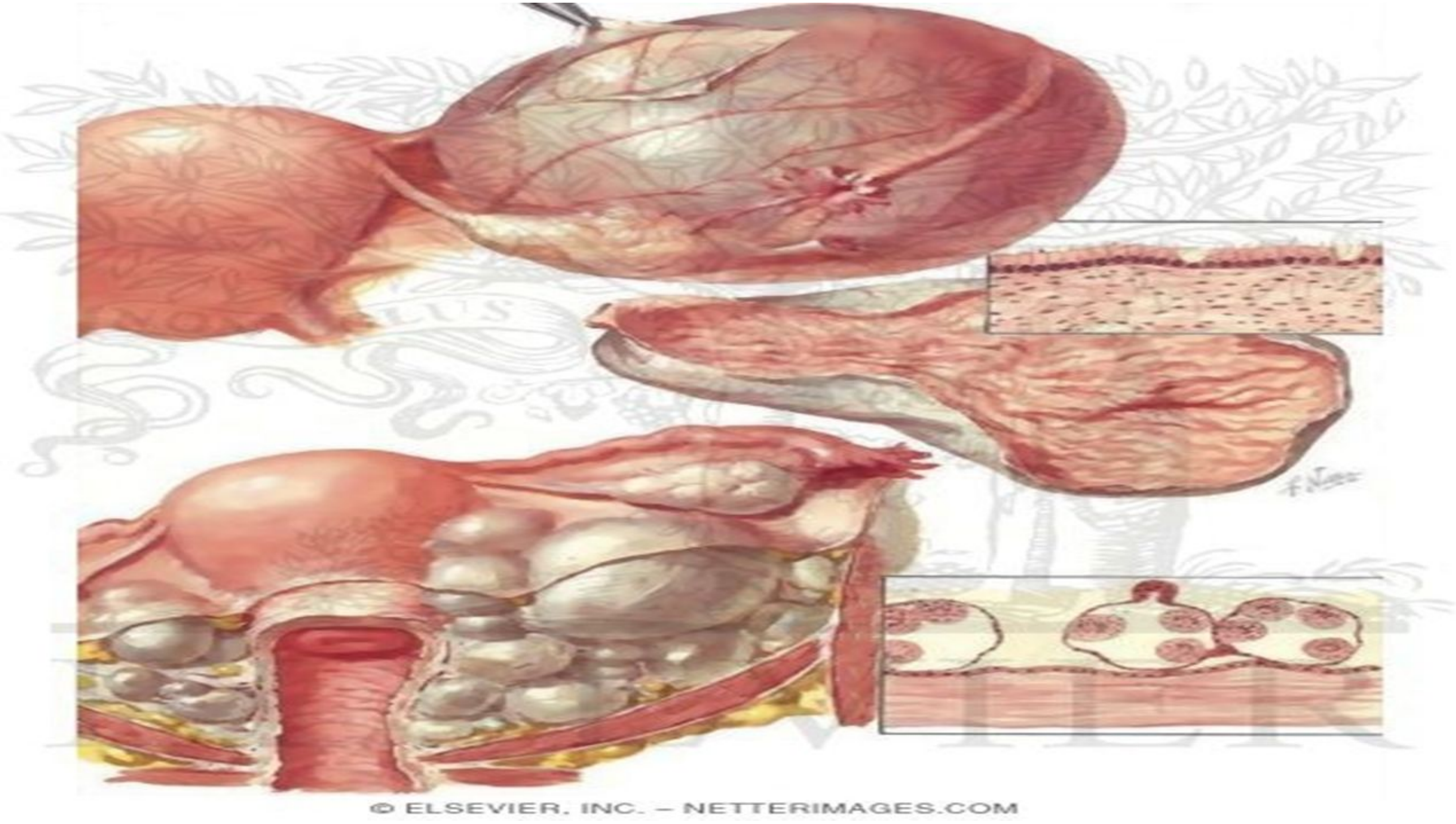
Vestigial structures :

- Vestigial remains of the mesonephric duct and tubules are always present in young children, but are variable structures in adults.

- The eoophoron, lies in the broad ligament between the mesovarium and the Fallopian tube.
- Paroophoron is situated in the broad ligament between the epoophoron and the uterus, a few rudimentary tubules are occasionally seen.
- In a few individuals, the caudal part of the mesonephric duct is well developed, running alongside the uterus to the internal os. This is the duct of Gartner.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